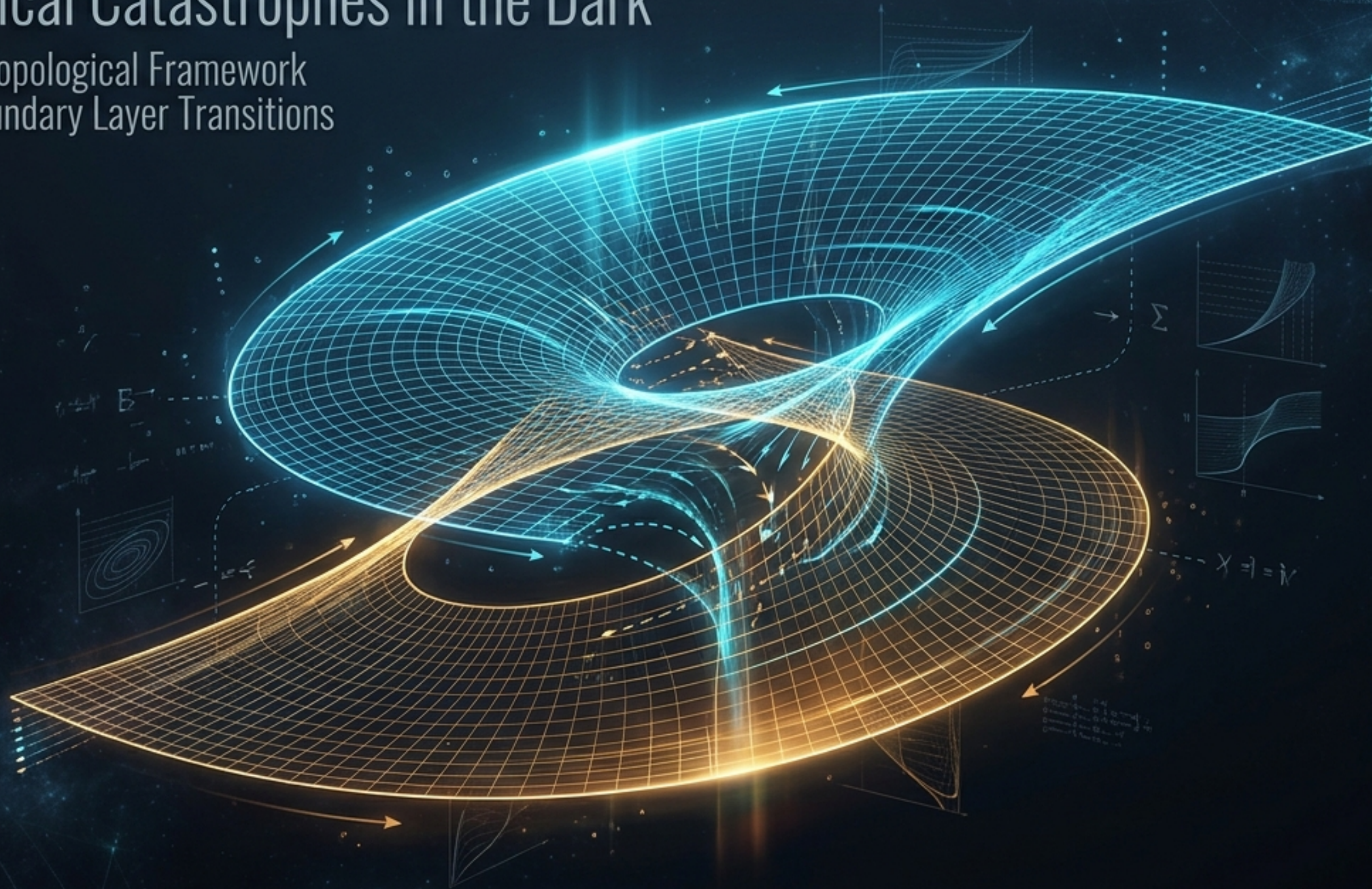
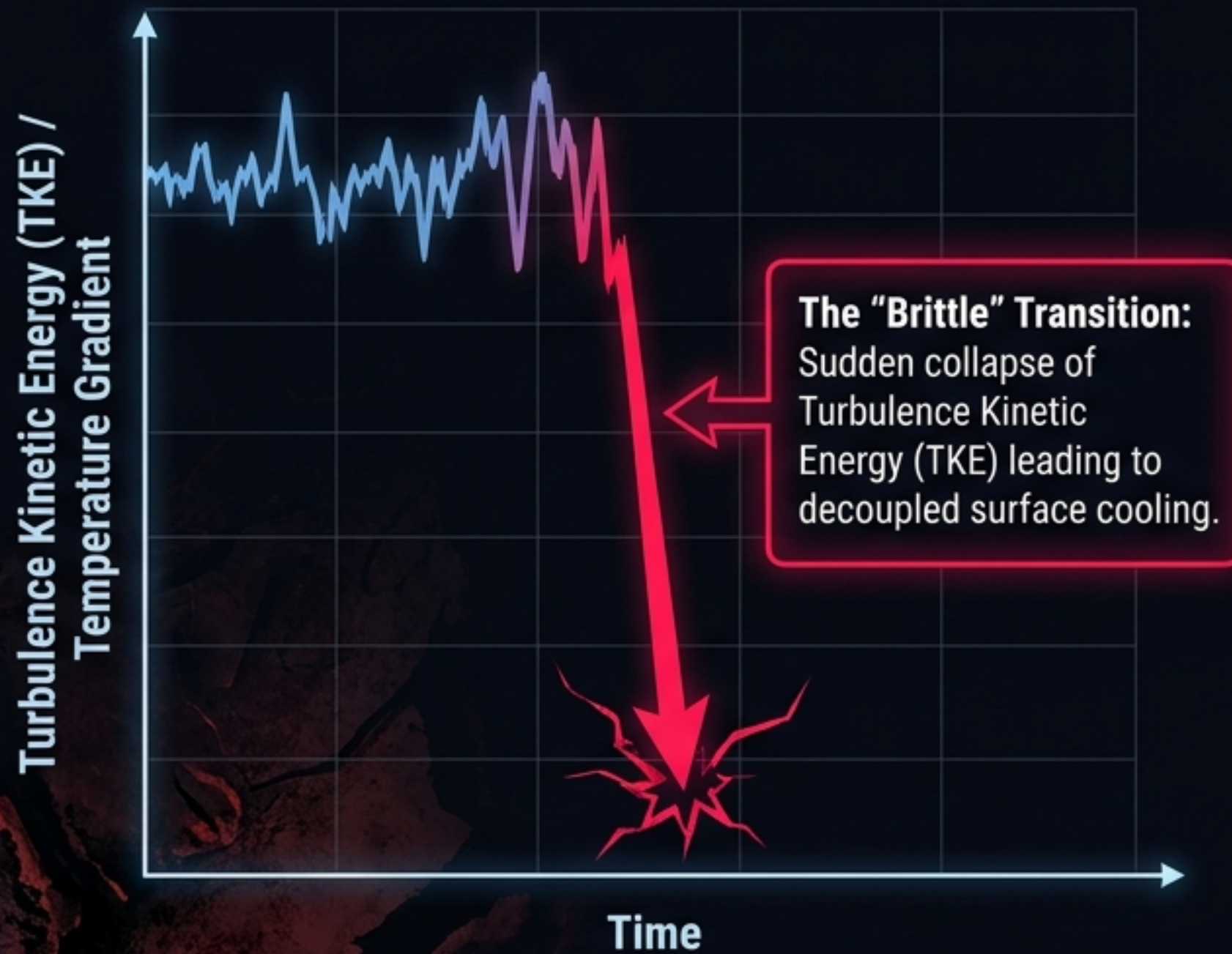


Geometrical Catastrophes in the Dark

A Fast-Slow Topological Framework
for Stable Boundary Layer Transitions



The Physical Reality



The Modeling Failure

$$Ri = \frac{g}{\theta} \frac{\partial \theta / \partial z}{(\partial u / \partial z)^2}$$

Mathematical Singularities:

Strongly stable conditions push Ri to critical values, causing division-by-zero errors and ill-conditioned Jacobians.

Unphysical Runaway Cooling:

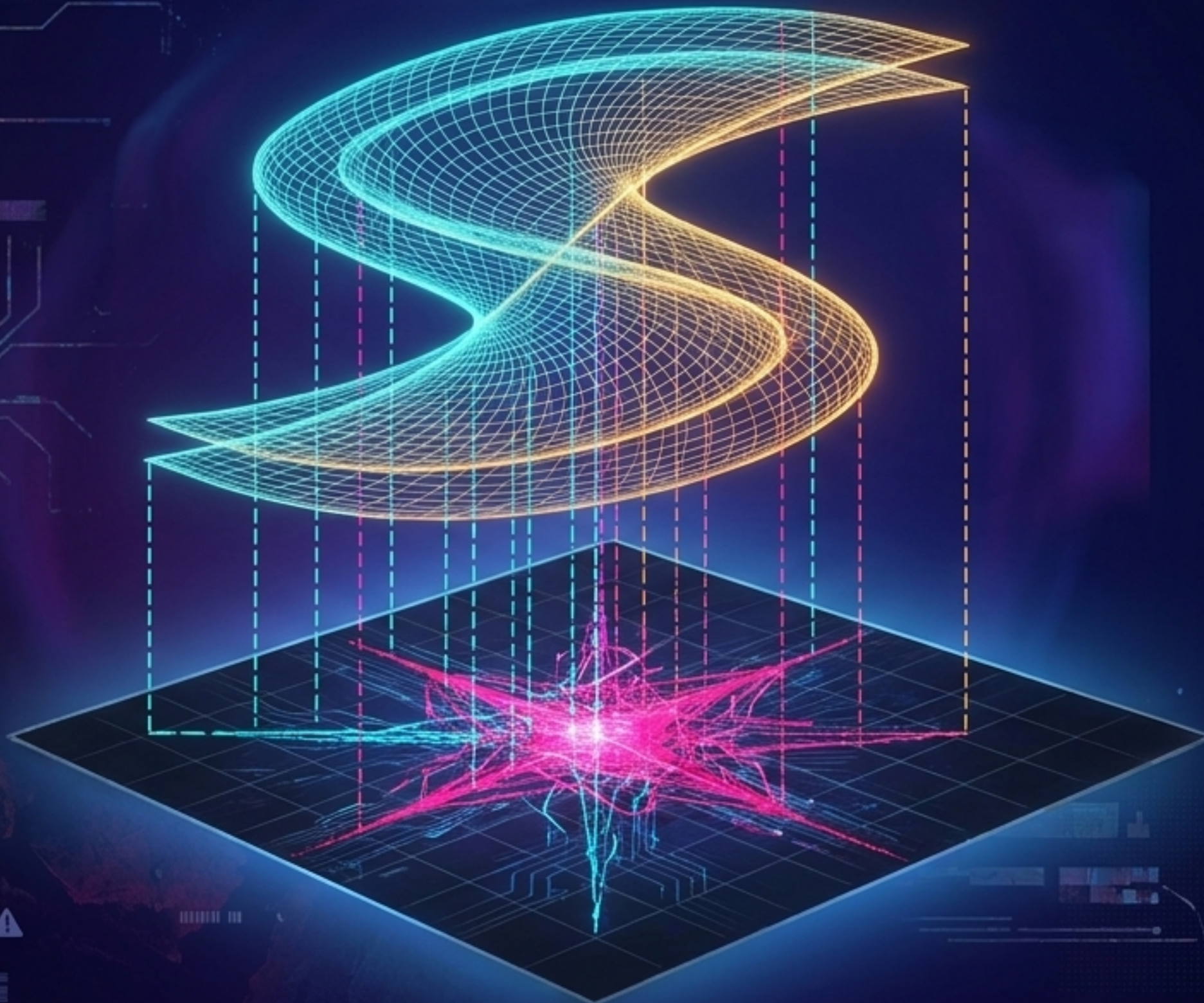
Models freeze up without arbitrary "long-tailed" stability functions.

Quasi-Steady Assumption Failure:

The SBL is inherently non-equilibrium. Turbulent adjustment is not infinitely fast.

We are patching numerical crashes by destroying the real physics.

The Root Cause: Projecting a Fold Onto a Flat Plane



Traditional MOST assumes a **single-valued** relationship between gradients and fluxes.

In **strongly stable** conditions, the system develops an **S-shaped slow manifold**—a complex, multi-valued state-space geometry.

MOST attempts to project this 3D fold onto a single flat coordinate axis. The numerical model crashes because it tries to find a middle ground that physically does not exist.

Architectural Upgrade: MOST vs. GSPT Framework

Monin-Obukhov Similarity Theory (MOST)

Quasi-steady equilibrium

Single-valued flat projection

Ad-hoc empirical stability functions

Mathematical singularity / division-by-zero

Arbitrary long-tailed smoothing

Geometric Singular Perturbation Theory (GSPT)

Non-equilibrium fast-slow dynamics

Multi-valued S-shaped manifold

Transversal boundary crossings and fold catastrophes

Catastrophic state jump

Rigorous structural regularization (δ)

Anatomy of the 4D Coriolis-Coupled System

Fast Turbulence Mechanics

$$\varepsilon \frac{de}{dt} = \sqrt{e + \delta} \left[\frac{\sigma}{h^2} (U^2 + V^2) - K \cdot G(T_s) - \alpha e \right]$$

Operates on a rapid timescale ($\varepsilon \ll 1$).
Dictates energetic allocation.

e : Turbulent Kinetic Energy

$$\gamma \sqrt{e + \delta}$$

Shear & Buoyancy
Feedbacks

Slow Macro-Environment

$$\begin{aligned} \frac{dU}{dt} &= -fV + \dots \\ \frac{dV}{dt} &= fU + \dots \\ \frac{dT_s}{dt} &= \dots \end{aligned}$$

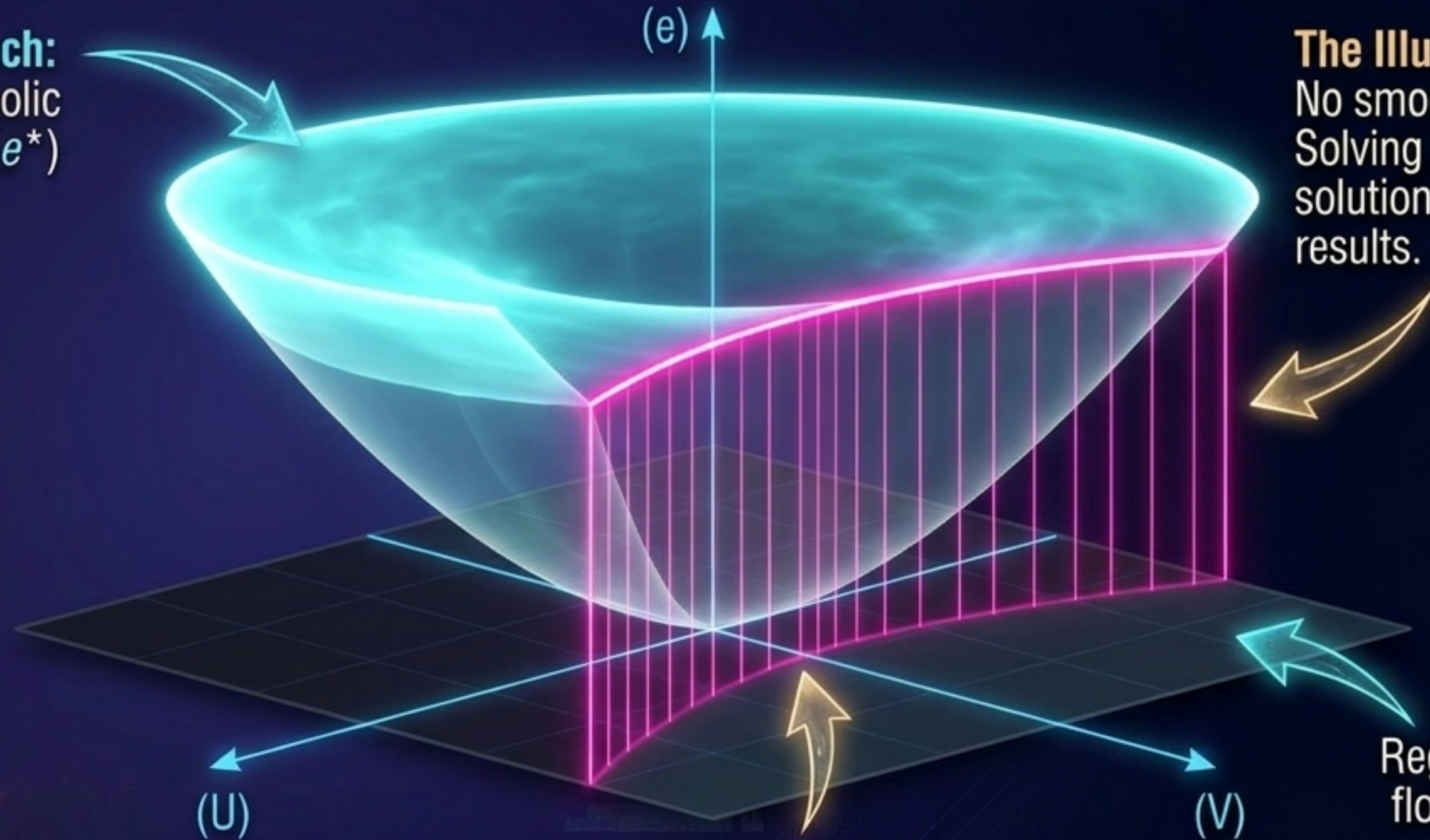
Operates on a slow thermodynamic/
rotational forcing timescale.

(U, V) : Horizontal Wind Momentum
 T_s : Surface Skin Temperature

Catastrophe Geometry: The Circular Elliptic Paraboloid

The Upper Branch:
Normally hyperbolic
turbulent sheet (e^*)

The Illusion of the Fold:
No smooth interior fold exists.
Solving for differentiable
solutions yields no real
results.



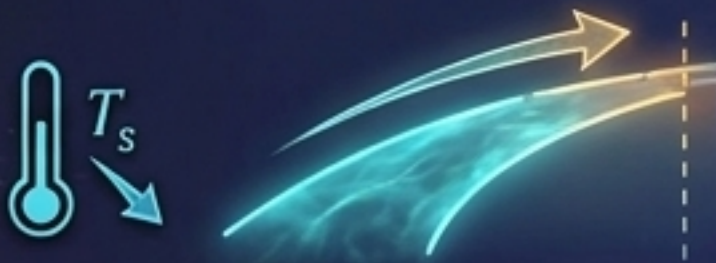
Transversal Crossing: The active branch
terminates cleanly at the physical boundary
($e=0$), forcing a sudden vertical plunge.

Regularized mixing
floor where $e \approx 0$

The Nocturnal Lifecycle as a Relaxation Oscillation

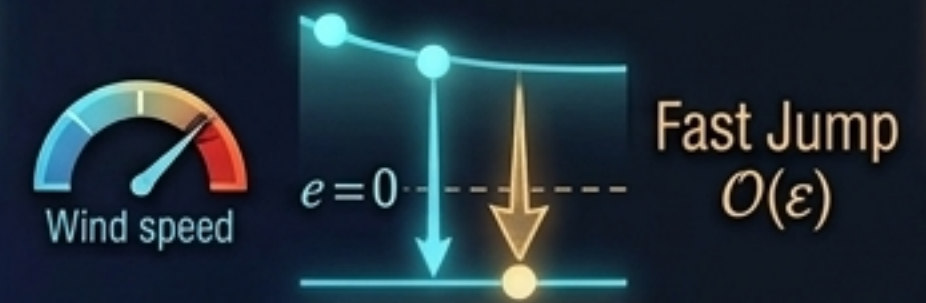
1. Evening Cooling (Slow)

Trajectory tracks the active branch. Radiative cooling drops T_{s1} shifting the manifold toward the crossing threshold.



2. Transversal Collapse (Fast Jump)

Wind speeds fall within the critical radius. The state crosses $e=0$ and plunges on a fast timescale $O(\epsilon)$ to the background floor.



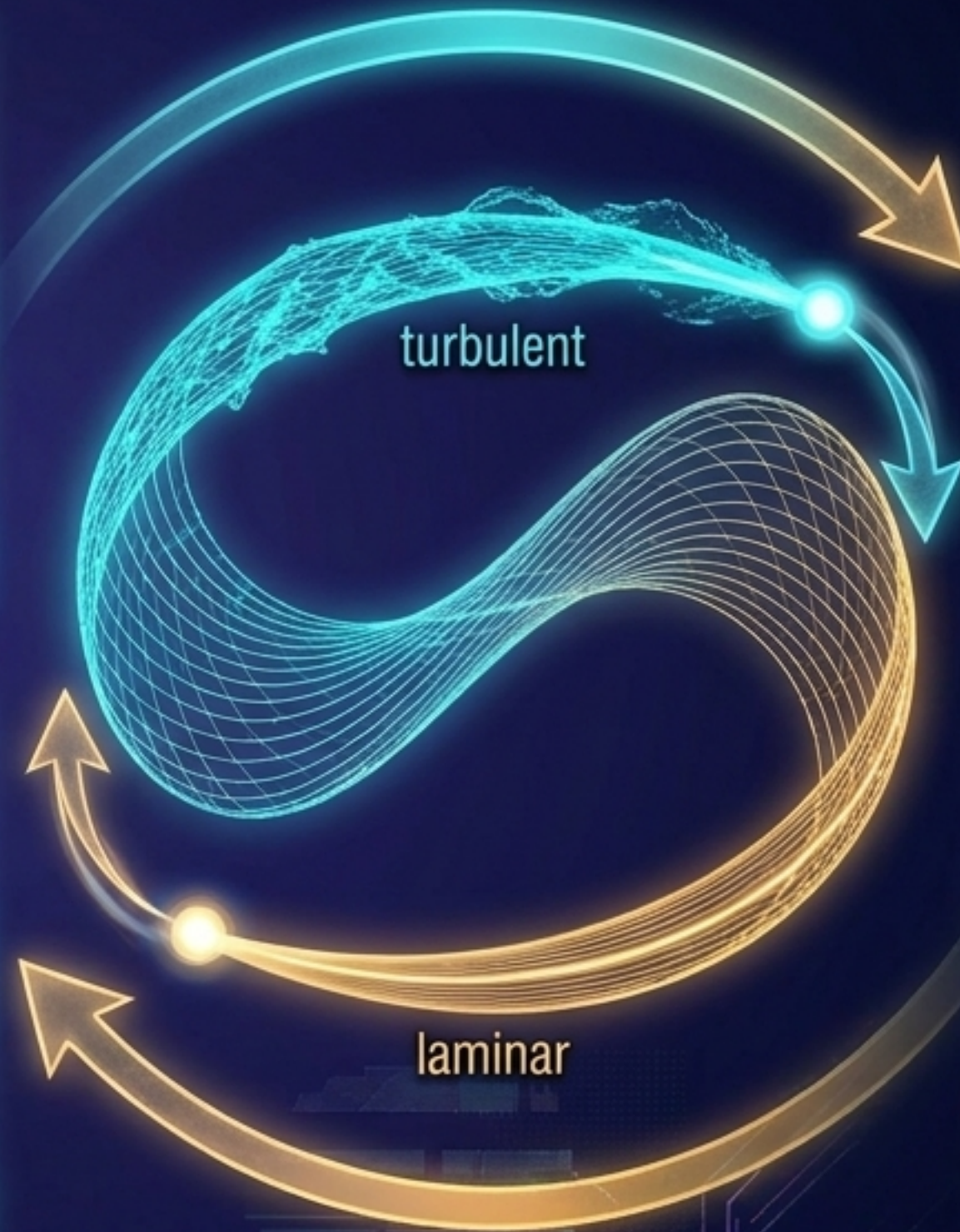
4. Turbulent Breakout (Fast Jump)

Mechanical shear production overcomes severe buoyancy. A fast-timescale jump back to the active turbulent canopy.



3. Inertial Acceleration (Slow)

Stripped of boundary layer friction, the unconstrained wind vector begins to accelerate.



The Blackadar Mechanism: Decoupled Inertial Orbits



When **brittle collapse** occurs, turbulent drag drops instantly (≈ 0).



- The decoupled wind vector executes broad inertial circuits at Coriolis frequency f .
- Re-ignition depends entirely on whether this expanding ageostrophic orbit generates enough mechanical vector shear to re-cross the 4D fold curve.

Mechanical Vector Shear



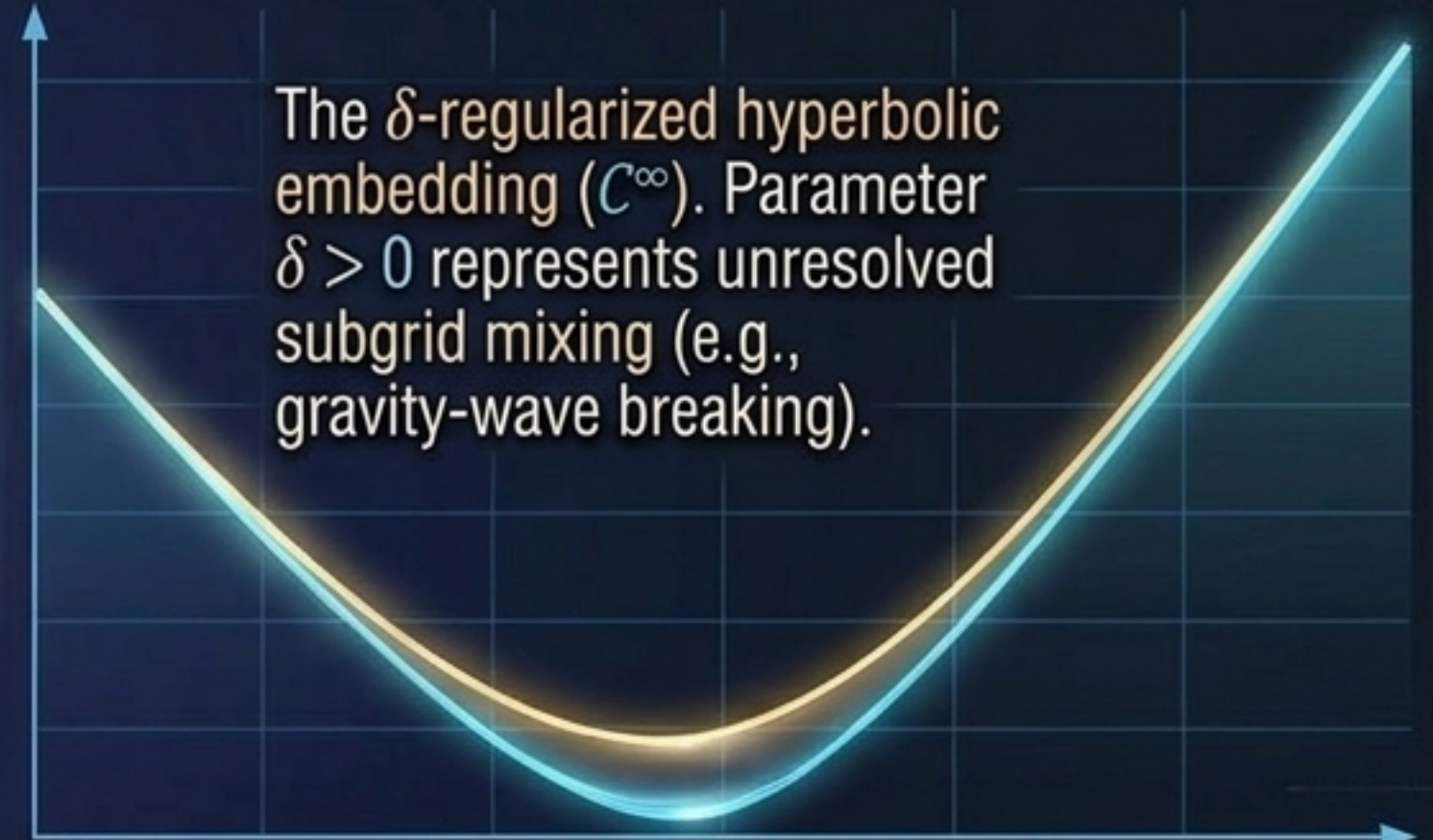
Structural Regularization: Taming the Finite-Time Singularity

Unregularized System



When production-buoyancy balance $\Delta < 0$, the fast subsystem hits a finite-time square-root singularity at $e = -\delta$. Differentiability is lost.

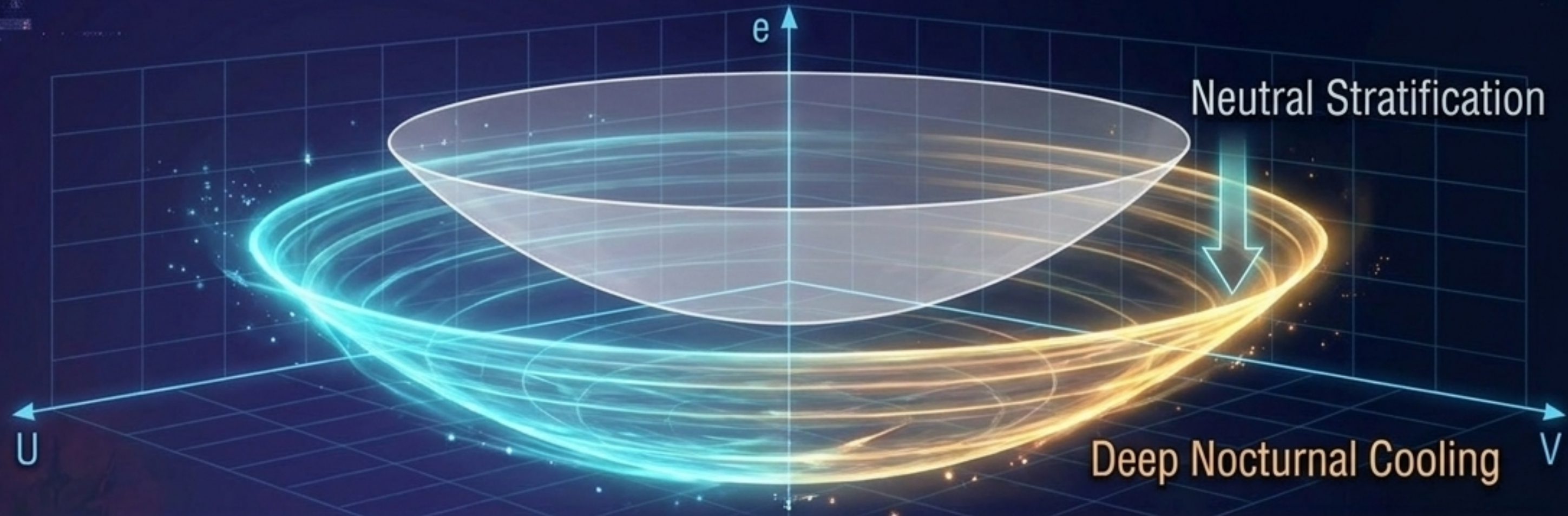
Regularized GSPT System



The δ -regularized hyperbolic embedding (C^∞). Parameter $\delta > 0$ represents unresolved subgrid mixing (e.g., gravity-wave breaking).

Result: Bounds eddy diffusivities safely away from zero, mathematically eliminating division-by-zero errors in implicit vertical diffusion solvers.

Thermal Shift Mechanics: The Asymmetry of Hysteresis



1. Unchecked radiative cooling drives skin temperatures down ($T_s \ll T_a$).



2. The stratification function $G(T_s)$ grows rapidly.



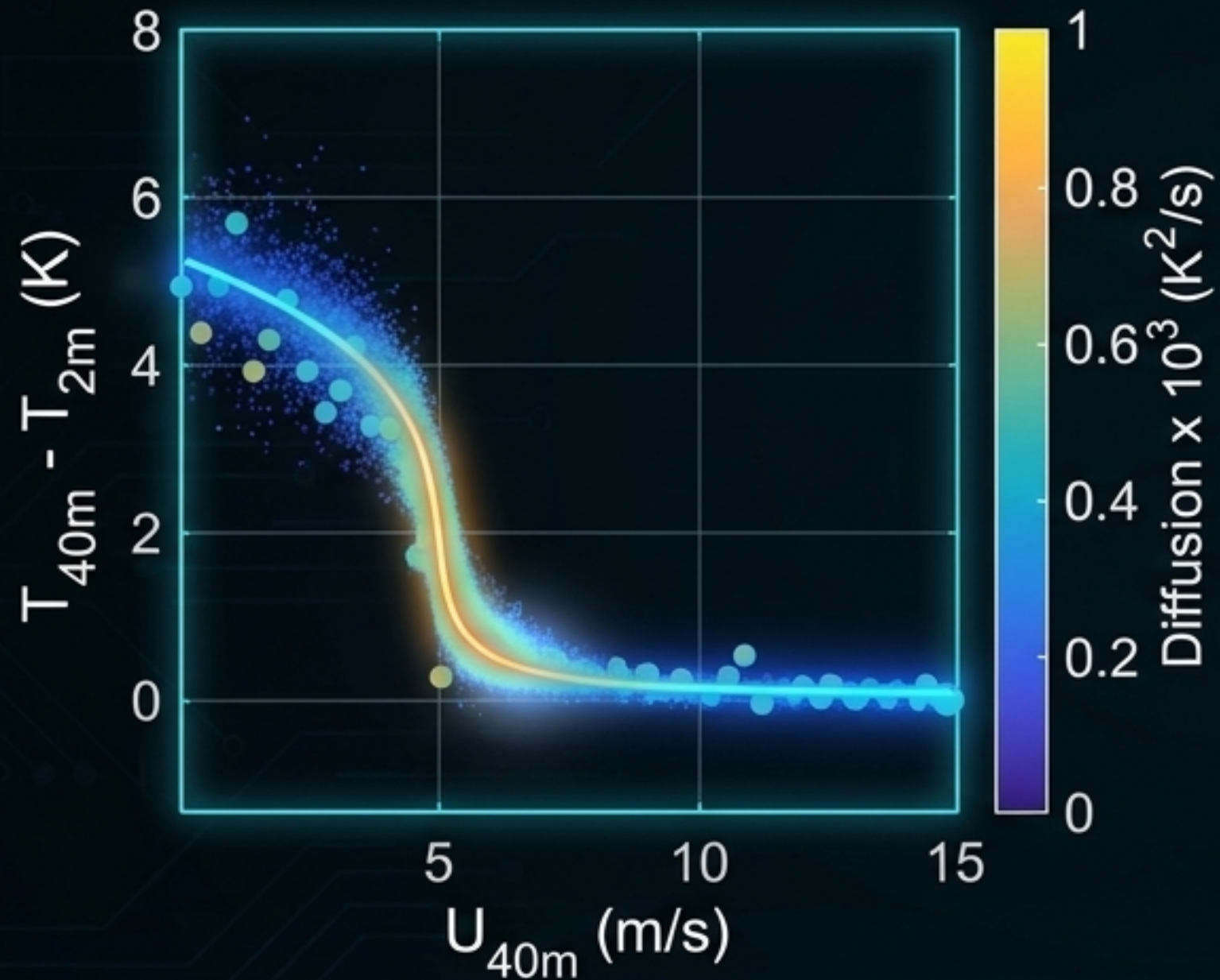
3. This forces the entire critical manifold downward, expanding the critical wind radius in the (U, V) plane required to maintain active mixing.



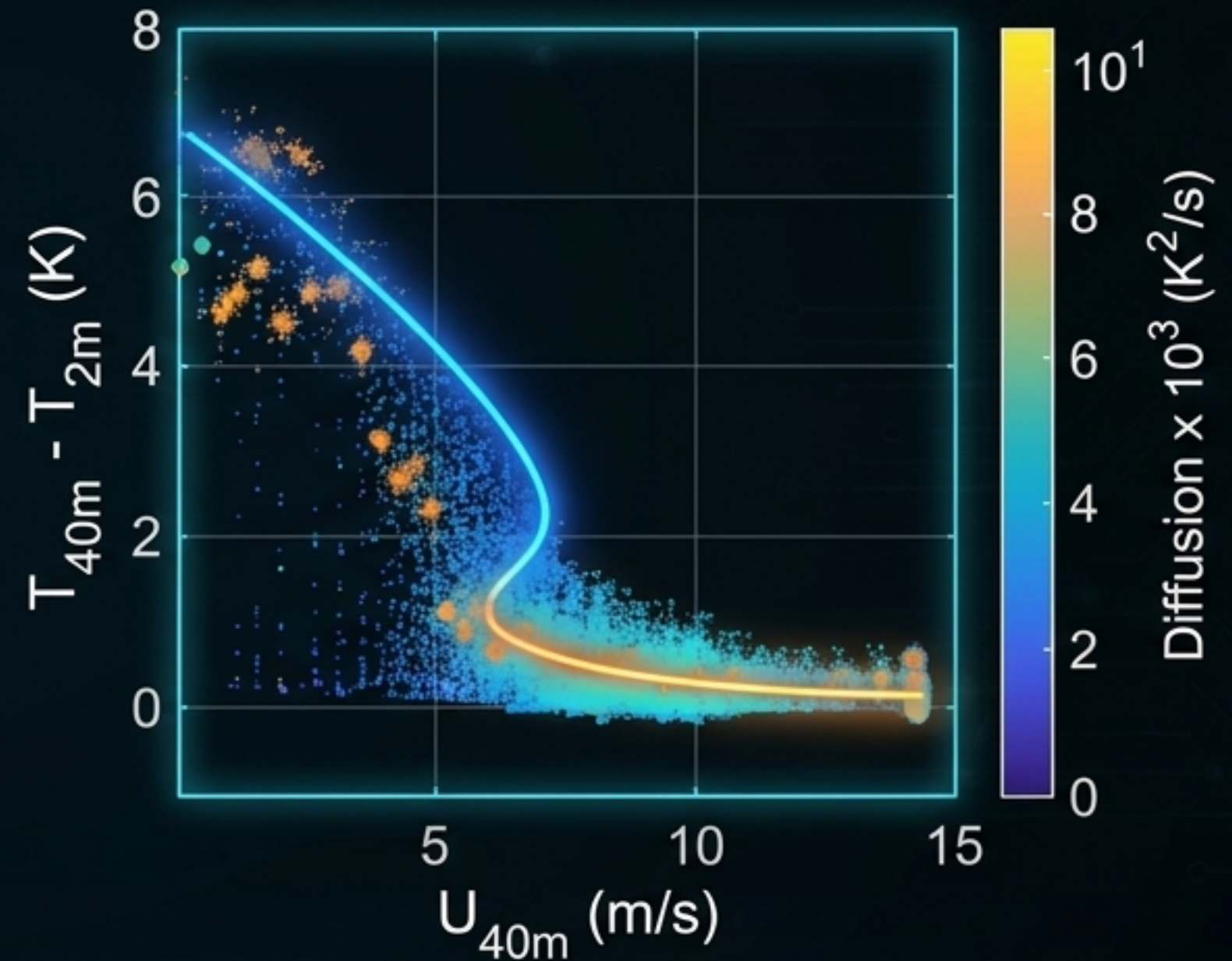
Result: The Low-Level Jet must generate significantly more shear to recover from the collapse than it needed to maintain turbulence initially.

Empirical Proof: Mapping Reality to the Manifold

Cabauw Tower, Netherlands



Dome C, Antarctica



3D state space (e , U , T_s) provides the **exact theoretical coordinate blueprint**. Real-world nocturnal transitions explicitly track the **catastrophic 'brittle' fold curves** predicted by GSPT.

Frictional Regularization Across Observational Regimes

Fluxes Over Snow Surface

Open Snow-Covered Landscape

RMSE = 0.3107



Low roughness snow surface provides minimal frictional damping, but strong stable stratification limits vertical mixing. Acts as a smooth, slippery boundary, allowing for unique surface flux dynamics and stable flows.

SHEBA

Perennial Polar Ice Sheet

RMSE = 1.1543



Radially pinned steady-state offset with minimal thermal boundary drift.

CASES99

Open Semi-Arid Grassland

RMSE = 1.2991



Low roughness provides minimal frictional damping. Following collapse, the decoupled wind vector detoremwever executes wide, unconstrained Blackadar inertial circuits, creating massive global deviations.

The three regimes demonstrate how **landscape-specific frictional damping** dictates the stability and **predictability** of the critical manifold. Higher roughness directly correlates with **tighter attractor pinning** and **lower RMSE**.

Mimetic Numerics: Preserving Topology at the Machine Level



Mapping to Differential Forms:

- 0-forms (Nodes): $\mathbf{T}_s$ (Localized intensive thermodynamic states)
- 1-forms (Edges): $\mathbf{U}, \mathbf{V}$ (Wind circulation metrics across layers)
- 2-forms (Volumes): $\mathbf{e}$ (Integrated TKE density)

The Mechanism:

DEC separates topological derivatives (d) from metric properties (Hodge Star $\star$).

The Result:

Eliminates spurious numerical grid shocks during rapid fast-timescale state jumps by mathematically guaranteeing **Kelvin's Circulation Theorem** holds **exactly** on the mesh.

A Unified Geometric Foundation



The atmospheric boundary layer is not a collection of isolated empirical adjustments. It is a **fundamental geometric manifold**. Standard vector calculus misses this; **topological dynamics** illuminate it.